The Hygiene Trap

The argument on one page

Measured: in ten homes, fridge-dispenser water carried thousands of times more bacteria than the tap feeding it, and no chlorine could be detected. **Our hypothesis:** over years that harms people, and bad breath may be one of the first signs. Every claim below carries a label.

Label	What the evidence shows	Source
ESTABLISHED	In ten homes in Tennessee, water from the fridge dispenser carried about 930 bacteria per mL against about 0.2 in the tap, and no chlorine could be detected. First-draw samples, one filter type, no health outcomes.	Swanson and He 2024
PLAUSIBLE	Drug-resistant bacteria were found on a hospital ice machine's spout and drain, and in patients. A fridge tap held *Mycobacterium abscessus* related to a cystic fibrosis patient's own; which way the bacteria travelled is unproven. One tap-water bacterium persisted in one person's gut for two weeks on bottled water.	Kanwar 2017; Thomson 2025; Lugli 2022
UNTESTED	Whether years of this water change a healthy person's gut, inflammation or ageing, and whether it contributes to bad breath through hydrogen sulfide, the tongue or the gut. No direct study found.	Our hypothesis
Cuts the other way	A 12-month infant trial found no overall change in the gut community from removing chlorine. Children on well water with far more bacteria had less allergy (observational; rural life is a confounder).	Parkin 2026; Turkalj 2020
REAL	Being judged, mistaken for a heavy drinker and passed over is real, whatever the cause. The imagined life in the video is not a study.	No study cited

What you can do

- Cold tap water in a clean jug or bottle is the simplest option.
- A jug filter is optional. It is a carbon filter too: change the cartridge on time, wash the jug, keep it cold.
- If you keep a dispenser: flush it first thing, change the filter on schedule, clean the spout and the drip tray.
- Buying a fridge: you can leave the dispenser out. Check whether the ice maker has its own water line.
- Symptoms that don't go away: see a doctor or a dentist. If anyone at home has a weak immune system, ask their doctor what is right for them.

How we will find out

1. Repeat the 2024 study in your own kitchen: paired first-draw samples, a chlorine test and a plate count (the replication protocol).
2. Do it in ten homes, with the same method, and share the results, good or bad.
3. Check a quick ATP test against the plate counts before anyone relies on it.
4. A blinded water swap with a breath meter, to see whether breath follows the fridge water.

Every step can say no. Decide the rules first, publish either way.

References

1. Swanson CS, He Q (2024). Linkage of in-refrigerator water dispensing systems to elevated exposure to microbial contamination at the microbiome scale. Am J Infect Control 52:857-859. PMID 38460731. doi:10.1016/j.ajic.2024.03.001
2. Kanwar A, Domitrovic TN, Koganti S, et al. (2017). A cold hard menace: A contaminated ice machine as a potential source for transmission of carbapenem-resistant Acinetobacter baumannii. Am J Infect Control 45:1273-1275. PMID 28625698. doi:10.1016/j.ajic.2017.05.007
3. Thomson RM, Wheeler N, Stockwell RE, et al. (2025). Infection by Clonally Related Mycobacterium abscessus Isolates: The Role of Drinking Water. Am J Respir Crit Care Med 211:842-853. PMID 40072241. doi:10.1164/rccm.202409-1824OC
4. Lugli GA, Longhi G, Mancabelli L, et al. (2022). Tap water as a natural vehicle for microorganisms shaping the human gut microbiome. Environ Microbiol 24:3912-3923. PMID 35355372. doi:10.1111/1462-2920.15988
5. Parkin K, Christophersen CT, Verhasselt V, et al. (2026). Chlorinated drinking water exposure enriches antimicrobial resistance pathways in the infant gut microbiome: a randomized trial. Commun Med (Lond) 6:405. PMID 42129350. doi:10.1038/s43856-026-01626-2
6. Turkalj M, Drkulec V, Haider S, et al. (2020). Association of bacterial load in drinking water and allergic diseases in childhood. Clin Exp Allergy 50:733-740. PMID 32270527. doi:10.1111/cea.13605

Source: The Hygiene Trap, print pages. https://fridge-water.ownhealth.bio/fridge-water#handouts (whole site: https://fridge-water.ownhealth.bio/). Printed November 6, 2026.
Every claim carries a label: established, plausible, untested, anecdote or real. This is a hypothesis, not a diagnosis: for persistent bad breath or symptoms, see a dentist and a doctor.
Text: CC BY 4.0. Please credit the Fridge water project, https://github.com/aptitudetechnology/fridge-water.